

Bhargav Hegde

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AI Engineer — Machine Learning Researcher — Autonomous Systems Specialist

Education

State University of New York at Buffalo, Buffalo, NY

Aug. 2024 – Present

Master of Science in Computer Science and Engineering

Research Associate - CAVAS Lab (Connected and Autonomous Vehicle Applications and Systems)

Research Focus: Real-time 3D perception, LiDAR-based object detection, and V2X cooperative perception for autonomous vehicles

JSS Academy of Technical Education, Bangalore, India

Aug. 2016 – Aug. 2020

Bachelor of Engineering in Computer Science and Engineering

Initiated and secured funding for an IoT laboratory; conducted instructional sessions for junior students

Professional Experience

Tech Mahindra, Software Engineer

Nov. 2021 – Jul. 2024

- Developed automation frameworks using Robot Framework with ML-based pattern recognition for Wi-Fi RTT locationing systems, achieving sub-meter accuracy in IEEE 802.11mc implementations
- Performed advanced packet analysis using Wireshark to optimize network protocols (DHCPv6, IPv6) and designed comprehensive test suites for stateful/stateless configurations
- Led R&D initiatives and delivered global training to cross-functional teams on Wi-Fi RTT technologies and implementation best practices

Tata Consultancy Services, System Associate Engineer

Apr. 2021 – Oct. 2021

- Developed Python-based RESTful APIs for automated business processes, streamlining data workflows for client reporting systems
- Accelerated deployment cycles by 15% through AWS EC2 testing environments and automated workflows

Pentagon Space, Intern

2020

- Completed intensive 4-month Python Full Stack development program covering Django, SQL, and web technologies
- Built scalable web applications using Python, Django, and SQL with focus on database optimization and RESTful API design

Teqed Labs, Intern

2019

- Developed Computer Vision-based Attendance System using Deep Learning for face recognition and automated database management
- **Awarded 'Best Project'** for delivering high-impact solution with real-time inference capabilities

Experts Hub, Intern

2018

- Built end-to-end ML pipeline for plant/seedling classification achieving 92% accuracy on 10,000+ image dataset
- Led team of 8 members in model development, data preprocessing, and deployment; **earned 'Best Intern' award**

Research & Selected Projects

DAIR-V2X Cooperative 3D Object Detection (Python, Shell, NVIDIA RTX 3080 Ti)

2025

- Full reproduction of DAIR-V2X late-fusion model for V2X cooperative perception, achieving 40.01% AP on vehicle detection
- Improved 3D AP@0.7 by 8.9% to 37.25% and BEV AP@0.7 by 6.7% to 50.51% while reducing communication overhead by 12.3%
- Implemented LiDAR point cloud processing with visualization tools for 3D and BEV fusion analysis
- GitHub: github.com/bhargavhegde/DAIR-V2X-Bhargav-2025

Autonomous Vehicle 3D Perception System (PyTorch, CUDA, ROS2, LiDAR)

Ongoing

- Designing real-time 3D object detection system using CenterPoint architecture optimized with PyTorch and CUDA
- Integrating Ouster LiDAR with ROS2 for real-time perception in dynamic autonomous driving environments
- Implementing efficient voxelization and sparse convolution techniques for low-latency inference

Low-Latency Sound Disambiguator (TensorFlow, Streamlit, Ollama + Mistral)

2024

- Real-time accessibility system for sound classification and directional detection using YAMNet and TDOA algorithms
- Implemented live microphone processing with AI-generated summaries, visual alerts, and polar compass interface
- Optimized for low-latency performance with confidence tracking and simulated stereo audio support
- GitHub: github.com/bhargavhegde/Low-latency-Sound-Disambiguator

RAG-KnowBot: Personal RAG Chatbot (Python, Ollama, LangChain, Chroma)

2024

- Built fully local Retrieval-Augmented Generation chatbot with Llama 3.1 8B and nomic-embed-text embeddings
- Implemented document processing (PDF, TXT, Markdown), vector database persistence, and hallucination guardrails
- Developed Streamlit interface with source citations, customizable prompts, and file management capabilities
- GitHub: github.com/bhargavhegde/RAG-KnowBot

Garbage Segregation System (Python, YOLOv3, CNN, Raspberry Pi)

Jun. 2020

- Developed robotic waste classification system achieving 90% accuracy using YOLOv3 and custom CNNs
- Integrated with robotic arm for automated sorting and placement; optimized edge inference on Raspberry Pi
- **Secured funding** from Karnataka State Council for Science and Technology (KSCST)

Advanced Machine Learning Implementations (PyTorch, Gymnasium)

2024

- Developed custom Gymnasium environment achieving 90%+ policy convergence with SARSA and n-step Double Q-learning
- Implemented CNNs from scratch achieving 87% accuracy; optimized VGG-13 and ResNet-34 architectures

Additional Projects

Laser Shooter with Object Detection: Real-time tracking system with OpenCV and servo control for precise laser targeting

Smart Kitchen Containers (IoT): Automated inventory monitoring with online ordering (Published in Matjournals)

Data Systems & Security: PySpark distributed processing handling 400,000+ words; network security analysis

Algorithm Analysis: Dynamic programming solutions achieving $O(n \log n)$ complexity for large-scale datasets

Technical Skills

Core Competencies

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv3, CNNs, ROS2, LangChain, Ollama

Programming Languages: Python, C/C++, Java, SQL, JavaScript, Shell Scripting, HTML/CSS

Deep Learning: Computer Vision, NLP, Reinforcement Learning (SARSA, Q-learning), LLMs (Llama, Mistral), RAG

Autonomous Systems: LiDAR Processing, 3D Object Detection, V2X Communications, Cooperative Perception

Cloud & DevOps: AWS (Lambda, S3, EC2), Azure, Docker, Kubernetes, Jenkins, CI/CD Pipelines, Git

Specialized Tools: CUDA, Spark, Optuna, Wireshark, Robot Framework, Chroma Vector DB, Streamlit

Databases & Systems: MySQL, SAP, Linux/UNIX, IoT, Embedded Systems (Raspberry Pi)

Networking: IEEE 802.11mc, DHCPv6, IPv6, Wi-Fi RTT, TLS/SSL, RESTful APIs

Relevant Coursework

Machine Learning, Analysis of Algorithms, Data-Intensive Computing, Computer Security, Database Systems, Modern Network Concepts, Computer Architecture, Operating Systems